



SYSTEM: 7ep

system: "7ep"

services:

- name: "Authentication Service"

responsibilities:

- "Register librarians with username and strong password"
- "Prevent duplicate librarian registration"
- "Authenticate librarian credentials"
- "Validate librarian sessions"
- "Securely handle credentials and denied access responses"

dependencies:

- "Audit Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Book Catalog Service"

responsibilities:

- "Register books by title without duplicates"
- "Delete registered books"
- "List registered books"
- "Consult books by identifier or title"
- "List books available for loan"

dependencies:

- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/book-catalog-service"

- name: "Reader Management Service"

responsibilities:

- "Register readers eligible for loans without duplicates"
- "Delete registered readers"
- "List registered readers"
- "Consult readers by identifier or name"

dependencies:

- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/reader-management-service"

- name: "Loan Management Service"

responsibilities:

- "Register book loans with loan dates"
- "Ensure books are not loaned to multiple readers simultaneously"
- "Validate reader and book data during loan registration"
- "Remove loan links when books or readers are deleted"
- "Preserve consistency among books readers and loans"

dependencies:

- "Book Catalog Service"
- "Reader Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/loan-management-service"

- name: "Audit Service"

responsibilities:

- "Record relevant operations from registration authentication consultation and loan flows"
- "Track system failures and operational behavior independently of business domains"

dependencies:

- ""

suggested_endpoints:

- "/api/audit-service"

- name: "Environment Management Service"

responsibilities:

- "Clear data and base structures to reset environment"
- "Recreate base structure"
- "Coordinate environment reset across functional areas to ensure consistent data state"

dependencies:

- "Authentication Service"
- "Book Catalog Service"
- "Reader Management Service"
- "Loan Management Service"

suggested_endpoints:

- "/api/environment-management-service"

- name: "Demo Utility Service"

responsibilities:

- "Perform addition of two integers"
- "Calculate Fibonacci sequence values using multiple approaches"
- "Calculate Ackermann function results using multiple approaches"
- "Validate mathematical inputs"
- "Provide service catalog for endpoint consultation and testing"

dependencies:

- ""

suggested_endpoints:

- "/api/demo-utility-service"

interactions:

- from: "Authentication Service"
to: "Audit Service"
- from: "Book Catalog Service"
to: "Loan Management Service"
- from: "Book Catalog Service"
to: "Audit Service"
- from: "Reader Management Service"
to: "Loan Management Service"
- from: "Reader Management Service"
to: "Audit Service"
- from: "Loan Management Service"
to: "Book Catalog Service"
- from: "Loan Management Service"
to: "Reader Management Service"
- from: "Loan Management Service"
to: "Audit Service"
- from: "Environment Management Service"
to: "Authentication Service"
- from: "Environment Management Service"
to: "Book Catalog Service"
- from: "Environment Management Service"
to: "Reader Management Service"
- from: "Environment Management Service"
to: "Loan Management Service"

SYSTEM: acmeair

system: "acmeair"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate customer credentials"
- "Create usage sessions"
- "Validate active customer sessions"
- "Invalidate sessions on logout"
- "Expire sessions automatically"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Flight Search Service"

responsibilities:

- "Search one-way and round-trip flights"
- "Manage registered airports and flight routes"

- "Provide flight departure and arrival schedules and reference fares"

dependencies:

- ""

suggested_endpoints:

- "/api/flight-search-service"

- name: "Booking Service"

responsibilities:

- "Create one-way and round-trip flight bookings"
- "Generate unique booking identifiers"
- "Record booking creation dates"
- "List and consult customer bookings"
- "Cancel existing bookings"

dependencies:

- "Authentication Service"
- "Flight Search Service"

suggested_endpoints:

- "/api/booking-service"

- name: "Customer Profile Service"

responsibilities:

- "Retrieve customer contact and address data"
- "Update customer contact and address data"
- "Confirm profile modifications with current password"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/customer-profile-service"

- name: "Loyalty Service"

responsibilities:

- "Present customer loyalty program status"
- "Display annual accumulated miles"
- "Display total accumulated miles"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/loyalty-service"

- name: "Environment Administration Service"

responsibilities:

- "Load and reload operational database with sample data"
- "Recreate operational records across services"
- "Provide operational indicators and record summaries"

dependencies:

- "Authentication Service"
- "Flight Search Service"

- "Booking Service"
- "Customer Profile Service"

suggested_endpoints:

- "/api/environment-administration-service"

interactions:

- from: "Booking Service"
to: "Authentication Service"
- from: "Booking Service"
to: "Flight Search Service"
- from: "Customer Profile Service"
to: "Authentication Service"
- from: "Loyalty Service"
to: "Authentication Service"
- from: "Environment Administration Service"
to: "Authentication Service"
- from: "Environment Administration Service"
to: "Flight Search Service"
- from: "Environment Administration Service"
to: "Booking Service"
- from: "Environment Administration Service"
to: "Customer Profile Service"

SYSTEM: Cargo-Tracker

system: "Cargo-Tracker"

services:

- name: "Cargo Management Service"

responsibilities:

- "Register cargo with origin"

dependencies:

- "destination"

suggested_endpoints:

- "/api/cargo-management-service"

- name: "Route Planning Service"

responsibilities:

- "Generate and present candidate routes matching cargo specifications"
- "Validate route legs"

dependencies:

- "load points"

suggested_endpoints:

- "/api/route-planning-service"

- name: "Cargo Tracking Service"

responsibilities:

- "Consolidate cargo operational status and full event history"

- "Calculate last known location"

dependencies:

- "transport status"

suggested_endpoints:

- "/api/cargo-tracking-service"

- name: "Logistics Event Registration Service"

responsibilities:

- "Receive and record logistics events including reception"

dependencies:

- "load"

suggested_endpoints:

- "/api/logistics-event-registration-service"

- name: "Integration Service"

responsibilities:

- "Ingest logistics events through operational interfaces"

dependencies:

- "integration services"

suggested_endpoints:

- "/api/integration-service"

interactions:

- from: "Cargo Management Service"

to: "destination"

- from: "Route Planning Service"

to: "load points"

- from: "Cargo Tracking Service"

to: "transport status"

- from: "Logistics Event Registration Service"

to: "load"

- from: "Integration Service"

to: "integration services"

SYSTEM: DayTrader7

system: "DayTrader7"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate investor credentials"
- "Start and manage active investor sessions"
- "End investor sessions through logout"
- "Enforce access control to restricted investor resources"

dependencies:

- "Investor Account Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Investor Account Service"

responsibilities:

- "Register new investors with initial balance"
- "Validate investor identification and password confirmation"
- "Maintain investor profile and financial balance"
- "Update investor profile data"
- "Record access metrics including login and logout counts"
- "Process balance debits and credits for trading operations"

dependencies:

- "Authentication Service"
- "Trading Service"
- "Portfolio Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/investor-account-service"

- name: "Market Data Service"

responsibilities:

- "Expose asset quotes for single and multiple asset queries"
- "Provide consolidated market summary and performance indices"
- "Update traded volume and market indicators after trades"
- "Handle invalid asset codes gracefully"

dependencies:

- "Trading Service"
- "Portfolio Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/market-data-service"

- name: "Trading Service"

responsibilities:

- "Generate and manage buy and sell trading orders"
- "Apply fixed operation fees to orders"
- "Track order lifecycle through status transitions"
- "Execute immediate or deferred order processing"
- "Prevent duplicate order execution and cancel invalid sales"
- "Maintain investor order history"
- "Deliver alerts for completed orders"

dependencies:

- "Investor Account Service"
- "Portfolio Service"
- "Market Data Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/trading-service"

- name: "Portfolio Service"

responsibilities:

- "Maintain investor asset positions"
- "Add positions resulting from completed purchases"
- "Remove positions resulting from completed sales"
- "Validate position availability for sale operations"
- "Calculate invested amount and market value and gain or loss"
- "Generate consolidated financial position summary"

dependencies:

- "Trading Service"
- "Market Data Service"
- "Investor Account Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/portfolio-service"

- name: "Simulation Administration Service"

responsibilities:

- "Configure simulation operational parameters"
- "Configure order processing modes and completed order alert settings"
- "Prepare and rebuild simulation database"
- "Reset simulation operational data"
- "Provide execution statistics and operational monitoring indicators"

dependencies:

- "Investor Account Service"
- "Trading Service"
- "Portfolio Service"
- "Market Data Service"

suggested_endpoints:

- "/api/simulation-administration-service"

interactions:

- from: "Authentication Service"
to: "Investor Account Service"
- from: "Investor Account Service"
to: "Authentication Service"
- from: "Investor Account Service"
to: "Trading Service"
- from: "Investor Account Service"
to: "Portfolio Service"
- from: "Investor Account Service"
to: "Simulation Administration Service"
- from: "Market Data Service"
to: "Trading Service"

- from: "Market Data Service"
to: "Portfolio Service"

- from: "Market Data Service"
to: "Simulation Administration Service"

- from: "Trading Service"
to: "Investor Account Service"

- from: "Trading Service"
to: "Portfolio Service"

- from: "Trading Service"
to: "Market Data Service"

- from: "Trading Service"
to: "Simulation Administration Service"

- from: "Portfolio Service"
to: "Trading Service"

- from: "Portfolio Service"
to: "Market Data Service"

- from: "Portfolio Service"
to: "Investor Account Service"

- from: "Portfolio Service"
to: "Simulation Administration Service"

- from: "Simulation Administration Service"
to: "Investor Account Service"

- from: "Simulation Administration Service"
to: "Trading Service"

- from: "Simulation Administration Service"
to: "Portfolio Service"

- from: "Simulation Administration Service"
to: "Market Data Service"

SYSTEM: Jokul

system: "Jokul"

services:

- name: "Authentication Service"

responsibilities:

- "Register users with credentials"
- "Authenticate users and manage active sessions"
- "Enforce profile-based access control for regular users and administrators"
- "Process user logout"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Catalog Service"

responsibilities:

- "List available movies with pagination"
- "Display movie details and cast information"
- "Manage movie categories and categorize movies"
- "Register new movies in the catalog"
- "Delete individual and batch movies from the catalog"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/catalog-service"

- name: "Media Service"

responsibilities:

- "Upload video files"
- "Manage video storage between local repository and delivery area"
- "Deliver video content for playback"
- "Provide download links for movie files"

dependencies:

- "Authentication Service"
- "Catalog Service"

suggested_endpoints:

- "/api/media-service"

interactions:

- from: "Catalog Service"
to: "Authentication Service"
- from: "Media Service"
to: "Authentication Service"
- from: "Media Service"
to: "Catalog Service"

SYSTEM: JpetStore

system: "JpetStore"

services:

- name: "Catalog Service"

responsibilities:

- "Organize products by animal categories"
- "Manage product and item details"
- "Search products by keyword"
- "Provide item information and stock status"

dependencies:

- "Inventory Service"

suggested_endpoints:

- "/api/catalog-service"

- name: "Customer Account Service"

responsibilities:

- "Register customer profile and access credentials"
- "Manage customer address and contact information"
- "Maintain customer preferences and favorite categories"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/customer-account-service"

- name: "Authentication Service"

responsibilities:

- "Authenticate customers with username and password"
- "Manage customer login sessions and logout"
- "Enforce access control for protected features"

dependencies:

- "Customer Account Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Shopping Cart Service"

responsibilities:

- "Manage temporary selection and quantities of cart items"
- "Consolidate duplicate items and recalculate quantities"
- "Calculate cart item totals and subtotal"
- "Clear cart upon request"

dependencies:

- "Catalog Service"
- "Inventory Service"

suggested_endpoints:

- "/api/shopping-cart-service"

- name: "Inventory Service"

responsibilities:

- "Track item stock availability and backorder status"
- "Provide updated stock consultations"
- "Register stock decrease upon completed purchase"

dependencies:

- "Catalog Service"

suggested_endpoints:

- "/api/inventory-service"

- name: "Order Service"

responsibilities:

- "Manage checkout workflow and order confirmation"
- "Process billing and delivery addresses and payment details"
- "Generate unique order identifier and record purchase details"
- "Provide order history and order detail consultation"

dependencies:

- "Authentication Service"
- "Customer Account Service"
- "Shopping Cart Service"
- "Inventory Service"

suggested_endpoints:

- "/api/order-service"

interactions:

- from: "Catalog Service"
to: "Inventory Service"
- from: "Customer Account Service"
to: "Authentication Service"
- from: "Authentication Service"
to: "Customer Account Service"
- from: "Shopping Cart Service"
to: "Catalog Service"
- from: "Shopping Cart Service"
to: "Inventory Service"
- from: "Inventory Service"
to: "Catalog Service"
- from: "Order Service"
to: "Authentication Service"
- from: "Order Service"
to: "Customer Account Service"
- from: "Order Service"
to: "Shopping Cart Service"
- from: "Order Service"
to: "Inventory Service"

SYSTEM: Pet-Clinic

system: "Pet-Clinic"

services:

- name: "Owner Service"

responsibilities:

- "Register owners"
- "Update owner information"
- "Search owners by last name"
- "Provide paginated owner list"
- "Display detailed owner profile"

dependencies:

- ""

suggested_endpoints:

- "/api/owner-service"

- name: "Animal Service"

responsibilities:

- "Register animals linked to owners"
- "Update animal details"
- "Validate animal birth date and unique name per owner"
- "Provide predefined list of animal types"

dependencies:

- "Owner Service"

suggested_endpoints:

- "/api/animal-service"

- name: "Visit Service"

responsibilities:

- "Register visits for animals"
- "Maintain animal visit history"
- "Record visit date and service description"
- "Resume pending visit processes after restart"

dependencies:

- "Animal Service"
- "Veterinarian Service"

suggested_endpoints:

- "/api/visit-service"

- name: "Veterinarian Service"

responsibilities:

- "List veterinarians and specialties"
- "Inform next expected visit for veterinarians"
- "Automatically assign visits prioritizing lowest workload"
- "Calculate and update veterinarian workload"
- "Provide veterinarian list in structured format"

dependencies:

- "Visit Service"

suggested_endpoints:

- "/api/veterinarian-service"

interactions:

- from: "Animal Service"

to: "Owner Service"

- from: "Visit Service"

to: "Animal Service"

- from: "Visit Service"

to: "Veterinarian Service"

- from: "Veterinarian Service"

to: "Visit Service"

SYSTEM: TNTConcept

system: "TNTConcept"

services:

- name: "Administration and Configuration Service"

responsibilities:

- "Maintain support records"
- "Manage user configurations"
- "Provide domain configuration parameters"

dependencies:

- "Identity and Access Service"
- "Workforce Management Service"
- "CRM and Proposals Service"
- "Billing Service"
- "Notification Service"
- "File Storage Service"

suggested_endpoints:

- "/api/administration-and-configuration-service"

- name: "Identity and Access Service"

responsibilities:

- "Register and maintain users"
- "Manage passwords and credentials"
- "Control user access profiles and scopes"
- "Validate user access"

dependencies:

- "Administration and Configuration Service"
- "Workforce Management Service"
- "Project and Work Tracking Service"
- "Content and Document Management Service"
- "Reporting Service"

suggested_endpoints:

- "/api/identity-and-access-service"

- name: "Workforce Management Service"

responsibilities:

- "Maintain work agreements"
- "Manage vacation requests"
- "Calculate vacation balances"
- "Process vacation approvals and expirations"

dependencies:

- "Administration and Configuration Service"
- "Identity and Access Service"
- "Notification Service"
- "Project and Work Tracking Service"
- "Commissioning Service"
- "Reporting Service"

suggested_endpoints:

- "/api/workforce-management-service"

- name: "Project and Work Tracking Service"

responsibilities:

- "Maintain client projects"
- "Register internal and external activities"
- "Classify activities and calculate worked hours"
- "Enforce evidence policies"

dependencies:

- "Identity and Access Service"
- "Workforce Management Service"
- "File Storage Service"
- "Notification Service"
- "CRM and Proposals Service"
- "Financial Management Service"
- "Billing Service"
- "Commissioning Service"
- "Reporting Service"

suggested_endpoints:

- "/api/project-and-work-tracking-service"

- name: "CRM and Proposals Service"

responsibilities:

- "Maintain organizations and contacts"
- "Record client interactions"
- "Manage proposals and proposal compositions"
- "Control proposal states"

dependencies:

- "Administration and Configuration Service"
- "Project and Work Tracking Service"
- "Billing Service"
- "Reporting Service"

suggested_endpoints:

- "/api/crm-and-proposals-service"

- name: "Financial Management Service"

responsibilities:

- "Maintain accounts and entries"
- "Calculate and track financial indicators"
- "Process financial entries and payments"

dependencies:

- "Project and Work Tracking Service"
- "Billing Service"
- "Reporting Service"

suggested_endpoints:

- "/api/financial-management-service"

- name: "Billing Service"

responsibilities:

- "Detail invoices and calculate totals"
- "Calculate taxes and balances"
- "Process proposal conversions"
- "Manage invoice attachments"

dependencies:

- "Administration and Configuration Service"
- "CRM and Proposals Service"
- "Project and Work Tracking Service"
- "Financial Management Service"
- "File Storage Service"
- "Reporting Service"

suggested_endpoints:

- "/api/billing-service"

- name: "Content and Document Management Service"

responsibilities:

- "Organize documents by category"
- "Manage document versions"
- "Associate documents with users and areas"

dependencies:

- "Identity and Access Service"
- "File Storage Service"
- "Reporting Service"

suggested_endpoints:

- "/api/content-and-document-management-service"

- name: "Commissioning Service"

responsibilities:

- "Register and manage orders and commissions"
- "Control commission workflows"

dependencies:

- "Project and Work Tracking Service"
- "Workforce Management Service"
- "File Storage Service"
- "Notification Service"
- "Reporting Service"

suggested_endpoints:

- "/api/commissioning-service"

- name: "Reporting Service"

responsibilities:

- "Consolidate cross-domain business data"
- "Generate and export reports"

dependencies:

- "Identity and Access Service"
- "Workforce Management Service"
- "Project and Work Tracking Service"
- "CRM and Proposals Service"
- "Financial Management Service"
- "Billing Service"
- "Content and Document Management Service"
- "Commissioning Service"

suggested_endpoints:

- "/api/reporting-service"

- name: "Notification Service"

responsibilities:

- "Send email notifications"
- "Process notification triggers"

dependencies:

- "Administration and Configuration Service"
- "Workforce Management Service"
- "Project and Work Tracking Service"
- "Commissioning Service"

suggested_endpoints:

- "/api/notification-service"

- name: "File Storage Service"

responsibilities:

- "Store files"
- "Retrieve files"

dependencies:

- "Administration and Configuration Service"
- "Project and Work Tracking Service"
- "Billing Service"
- "Content and Document Management Service"
- "Commissioning Service"

suggested_endpoints:

- "/api/file-storage-service"

interactions:

- from: "Administration and Configuration Service"
to: "Identity and Access Service"
- from: "Administration and Configuration Service"
to: "Workforce Management Service"
- from: "Administration and Configuration Service"
to: "CRM and Proposals Service"
- from: "Administration and Configuration Service"
to: "Billing Service"

- from: "Administration and Configuration Service"
to: "Notification Service"

- from: "Administration and Configuration Service"
to: "File Storage Service"

- from: "Identity and Access Service"
to: "Administration and Configuration Service"

- from: "Identity and Access Service"
to: "Workforce Management Service"

- from: "Identity and Access Service"
to: "Project and Work Tracking Service"

- from: "Identity and Access Service"
to: "Content and Document Management Service"

- from: "Identity and Access Service"
to: "Reporting Service"

- from: "Workforce Management Service"
to: "Administration and Configuration Service"

- from: "Workforce Management Service"
to: "Identity and Access Service"

- from: "Workforce Management Service"
to: "Notification Service"

- from: "Workforce Management Service"
to: "Project and Work Tracking Service"

- from: "Workforce Management Service"
to: "Commissioning Service"

- from: "Workforce Management Service"
to: "Reporting Service"

- from: "Project and Work Tracking Service"
to: "Identity and Access Service"

- from: "Project and Work Tracking Service"
to: "Workforce Management Service"

- from: "Project and Work Tracking Service"
to: "File Storage Service"

- from: "Project and Work Tracking Service"
to: "Notification Service"

- from: "Project and Work Tracking Service"
to: "CRM and Proposals Service"

- from: "Project and Work Tracking Service"
to: "Financial Management Service"

- from: "Project and Work Tracking Service"
to: "Billing Service"

- from: "Project and Work Tracking Service"
to: "Commissioning Service"

- from: "Project and Work Tracking Service"

to: "Reporting Service"

- from: "CRM and Proposals Service"

to: "Administration and Configuration Service"

- from: "CRM and Proposals Service"

to: "Project and Work Tracking Service"

- from: "CRM and Proposals Service"

to: "Billing Service"

- from: "CRM and Proposals Service"

to: "Reporting Service"

- from: "Financial Management Service"

to: "Project and Work Tracking Service"

- from: "Financial Management Service"

to: "Billing Service"

- from: "Financial Management Service"

to: "Reporting Service"

- from: "Billing Service"

to: "Administration and Configuration Service"

- from: "Billing Service"

to: "CRM and Proposals Service"

- from: "Billing Service"

to: "Project and Work Tracking Service"

- from: "Billing Service"

to: "Financial Management Service"

- from: "Billing Service"

to: "File Storage Service"

- from: "Billing Service"

to: "Reporting Service"

- from: "Content and Document Management Service"

to: "Identity and Access Service"

- from: "Content and Document Management Service"

to: "File Storage Service"

- from: "Content and Document Management Service"

to: "Reporting Service"

- from: "Commissioning Service"

to: "Project and Work Tracking Service"

- from: "Commissioning Service"

to: "Workforce Management Service"

- from: "Commissioning Service"

to: "File Storage Service"

- from: "Commissioning Service"

to: "Notification Service"

- from: "Commissioning Service"

to: "Reporting Service"

- from: "Reporting Service"
to: "Identity and Access Service"

- from: "Reporting Service"
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to: "Project and Work Tracking Service"

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- from: "File Storage Service"
to: "Commissioning Service"